COLUMBIA UNIVERSITY

DEPARTMENT OF BIOSTATISTICS

P8100 – APPLIED REGRESSION I

Exercise Sheet 3 – Model Answers

Question 1 [2+2+2+2 = 8 marks]

(a) For the *i*th subject, let Y_i be the final phosphate measurement, X_i be the initial phosphate measurement, and Z_i be the group belonging, where

$$Z_i = \begin{cases} 1, & \text{obese group} \\ 0, & \text{control group} \end{cases}$$

Then the ANCOVA model is

$$\mathrm{E}\left(Y_i \mid X_i, Z_i\right) = \beta_0 + \beta_1 X_i + \beta_2 Z_i$$ [a]

(b) Fitting the model in (a) results in:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval		Hypothesis Test		
			Lower	Upper	Wald Chi-Square	df	Sig.
(Intercept)	.889	.6076	-.302	2.080	2.140	1	.144
Initial	.600	.1437	.319	.882	17.450	1	<.001
[group=1]	-.234	.2054	-.636	.169	1.298	1	.255
[group=0]	0[a]	.	.	.	.	.	.
(Scale)	.301[b]	.0741	.186	.488			

Dependent Variable: final
Model: (Intercept), Initial, group

a. Set to zero because this parameter is redundant.
b. Maximum likelihood estimate.

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[a] This can also be written as $Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \varepsilon_i$

The p-value for ‘group’ is .225 > .05, implying that there is no significant difference in the mean outcome between the two groups.

(c) Yes, the initial phosphate value contributes significantly to the final phosphate value. Each unit increase in the initial value is significantly ($p<.001$) associated with a .600 increase in the mean final value.

(d) The model fitted now is:

$$\mathrm{E}\left(Y_i \mid X_i, Z_i\right) = \beta_0 + \beta_1 X_i + \beta_2 Z_i + \beta_3 X_i Z_i$$

with output:

Parameter Estimates

Parameter	B	Std. Error	95% Wald Confidence Interval		Hypothesis Test		
			Lower	Upper	Wald Chi-Square	df	Sig.
(Intercept)	.538	1.4840	-2.371	3.447	.131	1	.717
Initial	.686	.3607	-.021	1.393	3.619	1	.057
[group=1]	.190	1.6493	-3.042	3.423	.013	1	.908
[group=0]	0[a]	.	.	.	.	.	.
[group=1] * Initial	-.102	.3932	-.873	.669	.067	1	.796
[group=0] * Initial	0[a]	.	.	.	.	.	.
(Scale)	.300[b]	.0740	.185	.487			

Dependent Variable: final
Model: (Intercept), Initial, group, group * Initial

a. Set to zero because this parameter is redundant.

b. Maximum likelihood estimate.

The output above shows that the ‘initial’ x ‘group’ interaction is not significant at the 5% level (pval = .795 > .05), i.e. the association between the initial and final phosphate values does not depend on the group.

Question 2 [2+2+2+2+2+2 = 12 marks]

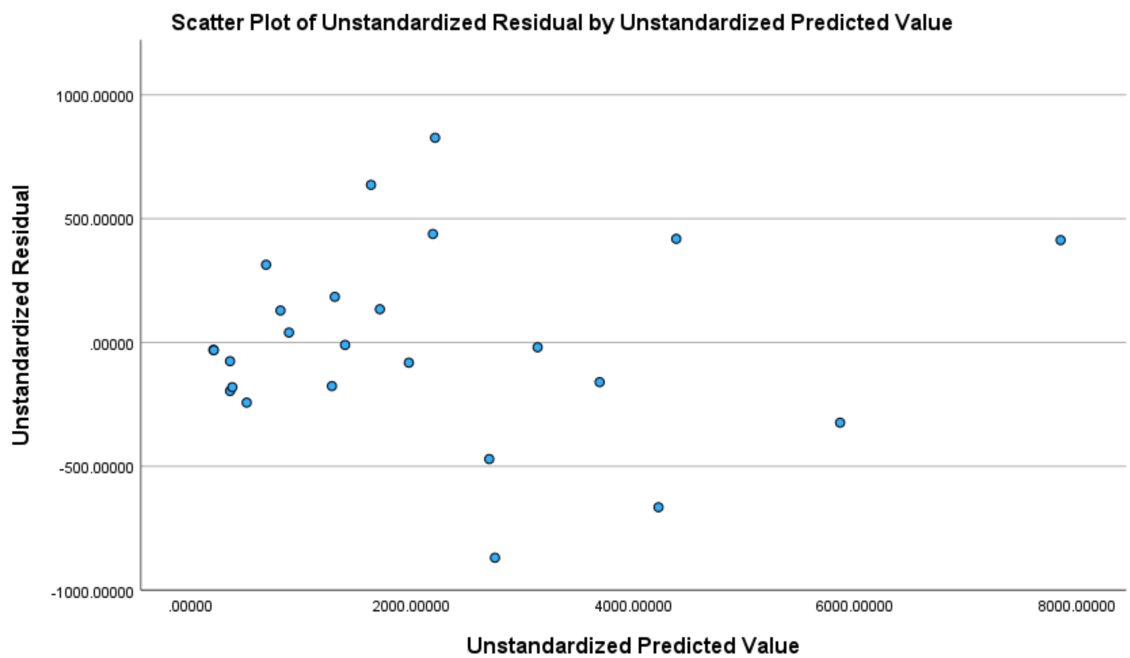
(a) The beta coefficients for the line of best fit is:

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	135.038	237.800		.568	.578
	x1	-1.284	.805	-.112	-1.596	.129
	x2	1.804	.516	.355	3.494	.003
	x3	.669	1.846	.020	.362	.722
	x4	-21.436	10.170	-.154	-2.108	.050
	x5	5.622	14.745	.035	.381	.708
	x6	-14.490	4.220	-.999	-3.434	.003
	x7	29.335	6.364	1.756	4.609	<.001

a. Dependent Variable: y

(b) The residual plot is:



The residual plot shows a funneling out effect (increase in variance with fitted value). This suggests the outcome should be log transformed[b].

(c) The model to be fitted is now:

$$\log Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_7 X_7 + \varepsilon$$

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	5.290	.290		18.235	<.001
	x1	-9.815E-5	.001	-.014	-.100	.922
	x2	.001	.001	.184	.895	.383
	x3	.007	.002	.331	2.936	.009
	x4	.019	.012	.221	1.497	.153
	x5	-.008	.018	-.087	-.470	.644
	x6	-.005	.005	-.568	-.963	.349
	x7	.010	.008	.997	1.290	.214

a. Dependent Variable: logy

(d) The required diagnostics are:

[b] A square root transformation is also possible,

Site	hat_diaginals	Standardized	Cook	Studentized
1	.21729	-.35445	.00544	-.34515
2	.12088	-.76249	.01393	-.75271
3	.12142	-1.06200	.02714	-1.06627
4	.12311	-1.58704	.06136	-1.66818
5	.10748	-1.01299	.02219	-1.01382
6	.11890	-1.79724	.07628	-1.93732
7	.14287	2.38388	.15898	2.83450
8	.31911	.93976	.06186	.93634
9	.24082	.97224	.04614	.97059
10	.08954	.24586	.00112	.23894
11	.08420	.78034	.01079	.77097
12	.16243	.86746	.02387	.86083
13	.04021	.40772	.00181	.39749
14	.05692	.56078	.00422	.54914
15	.51776	-.37604	.02229	-.36634
16	.36234	-.08966	.00068	-.08700
17	.32829	-.25696	.00481	-.24977
18	.40645	-.61992	.03874	-.60832
19	.04681	.75462	.00677	.74466
20	.32619	-.29944	.00648	-.29127
21	.03040	.99094	.00930	.99038
22	.74536	-1.39054	.88437	-1.43297
23	.94846	-.94720	9.60631	-.94418
24	.83618	.16721	.02473	.16235
25	.50660	-.21430	.00692	-.20818

(e) Based on Cook's distance, the point associated with site 23 is influential[c]. The hat diagonal for this point is .94846 (close to unity) suggesting that its distance from the majority of *X*-points contributes substantively to its influence. Its standardized residual is only moderately large suggesting the vertical deviation of this point does not have a substantive contribution to its influence.

(f) From t-tables, the 5% cut-off value is $t_{16;.05} = 2.1199$ and the point associated with site 7 is an outlier[d]. This point is now removed the model

$$\log Y = \beta_0 + \beta_1 X_1 + \ldots + \beta_7 X_7 + \varepsilon$$

[c] If a square root transformation was used, then there may be other influential points. Check data points with $D_i > 1$

[d] Again, if a square root transformation was used, then there may be other outliers. Check data points with Studentized $t > 2.1199$

is fitted again, resulting in

Coefficients[a]

Model		Unstandardized Coefficients B	Unstandardized Coefficients Std. Error	Standardized Coefficients Beta	t	Sig.
1	(Constant)	4.993	.266		18.805	<.001
	x1	-1.292E-5	.001	-.002	-.016	.988
	x2	.001	.001	.226	1.322	.205
	x3	.008	.002	.385	4.144	<.001
	x4	.024	.011	.278	2.260	.038
	x5	-.008	.015	-.077	-.498	.625
	x6	-.004	.004	-.455	-.926	.368
	x7	.008	.007	.786	1.218	.241

a. Dependent Variable: logy